Module Review: Bacterial Purine Nucleoside & Nucleobase Salvage in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI:txid160488; proteome UP000000556) **Target bucket:** KEGG `ppu00230` "Purine metabolism" (broad overview map) **Module area:** nucleotide_metabolism **Purpose:** species-aware curation review supporting manual module satisfiability and gene-annotation triage. Curation verdicts use the controlled vocabulary: `covered`, `candidate_uncertain`, `gap`, `not_expected_in_target_taxon`, `module_needs_revision`.

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1. Executive summary

The commissioned module — *phosphorolysis of purine nucleosides followed by re-fixation of the free bases to nucleotides* — is a **narrow 4-step process**, whereas the 65-gene candidate list is a projection of the entire KEGG overview map `ppu00230` plus ~15 neighboring buckets. **Only 4 of the 65 candidates encode true salvage steps:** `ppnP` (PP_4248), `apt` (PP_4266), `hpt` (PP_0747) and `xpt` (PP_5265). The other 61 belong to de novo IMP biosynthesis, nucleotide interconversion/kinases, purine **catabolism**/ureide nitrogen assimilation, housekeeping nucleotidases/Nudix hydrolases, and nucleotide-based signalling, and should be scored as neighboring-pathway context, not salvage evidence.

The module is **nominally satisfiable**. The three phosphoribosyltransferase (PRT) branches are all present: - **Adenine → AMP** (`apt`/PP_4266) — `covered`, high-quality (SwissProt-reviewed, EC 2.4.2.7). - **Xanthine → XMP** (`xpt`/PP_5265) — `covered`, reviewed (EC 2.4.2.22). - **Hypoxanthine/Guanine → IMP/GMP** (`hpt`/PP_0747) — `covered` but **`candidate_uncertain`** (unreviewed, "Predicted", no EC assigned).

The **one substantive weakness is step 1 (nucleoside phosphorolysis).** A proteome-wide UniProt search shows that the **only** enzyme in KT2440 bearing EC 2.4.2.1 is the accessory Cupin-fold **PpnP** (PP_4248); there is **no DeoD (PNP-I) or XapA (PNP-II) ortholog** — the canonical high-flux bacterial purine nucleoside phosphorylases. The best uncharacterised candidate for a classical phosphorylase is **PP_3254** (PF01048 / IPR000845 "nucleoside phosphorylase domain"), which is **absent from the candidate list**. `yfiH`/PP_0624, listed as "purine nucleoside phosphorylase", is an **unreviewed, EC-less, likely over-propagated annotation** (YfiH/DUF152 polyphenol-oxidase-like family) and should not be counted.

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2. Target-organism pathway definition

Included (this module): the salvage arm only — 1. Phosphorolytic cleavage of purine ribonucleosides (adenosine, inosine, guanosine, xanthosine) → free base + α-D-ribose-1-phosphate; 2. Adenine + PRPP → AMP + PPi (APRT); 3. Hypoxanthine/Guanine + PRPP → IMP/GMP + PPi (HGPRT); 4. Xanthine + PRPP → XMP + PPi (XPRT).

All three PRT branches consume **PRPP** (supplied by `prs`/PP_0722 and the second PRPP synthase PP_2744) and release **diphosphate**; this is a boundary input, not a salvage step.

Explicitly kept separate (neighboring maps / must not be scored as salvage): - **De novo IMP biosynthesis** (`purF, purD, purN, purT, purL, purM, purK, purE, purC, purB, purH`) and the IMP→AMP/GMP branch (`purA, purB, guaA, guaB`). - **Nucleotide interconversion / kinases** (`adk, gmk, ndk, nrdAB`, `prs`, `PP_2744`). - **Purine catabolism → ureide/nitrogen assimilation** (`xdhAB` xanthine dehydrogenase; `pucL, pucM, puuE, allA, allE, PP_4310`; `ureABC`). This is a *distinct* degradative pathway (purine → urate → allantoin → glyoxylate + NH_3/urea) and in *P. putida* underlies purine/allantoin use as N-source. It should be its **own module**, not folded into salvage. - **Molybdo-hydroxylase mis-mapping:** `paoABC` (PP_3308–3310), annotated "promiscuous aromatic **aldehyde** dehydrogenase" (EC 1.2.99.7), is a PaoABC-type aldehyde oxidoreductase erroneously projected onto the purine map — **not** a purine enzyme. - **Housekeeping nucleotidases / Nudix / NTP-sanitising** (`yrfG, surE, ushA, PP_2531, nudE, nudF, mazG, apaH, PP_5100, dgt`) and **nucleotide signalling** (`relA, spoT, cyaA, pde, ppx`). - **Cross-listed non-purine enzymes**: `cysD/cysNC` (ATP sulfurylase), `arcC` (carbamate kinase), `pgm/cpsG/algC` (phospho-mutases).

Alternate names / DB definitions: KEGG `ppu00230` = "Purine metabolism" (overview, not a module). MetaCyc/BioCyc split the equivalent biology into "purine ribonucleosides degradation" / "purine nucleobases salvage" / "adenine and adenosine salvage" / "guanine and guanosine salvage". The relevant GO umbrella is **purine-containing compound salvage (GO:0043101)** with base-specific children (see §6).

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3. Expected step model

#	Step	Reaction (EC)	Expected enzyme family	KT2440 candidate
1	Nucleoside phosphorolysis	nucleoside + Pi → base + R1P (2.4.2.1/2.4.2.2)	PNP-I (DeoD)/PNP-II (XapA)/PpnP-Cupin	`ppnP` PP_4248 (+ candidate PP_3254)
2	Adenine salvage	adenine + PRPP → AMP (2.4.2.7)	APRT (PF00156)	`apt` PP_4266
3	Hypoxanthine/ guanine salvage	Hx/Gua + PRPP → IMP/ GMP (2.4.2.8)	HGPRT (PF00156)	`hpt` PP_0747
4	Xanthine salvage	xanthine + PRPP → XMP (2.4.2.22)	XPRT (PF00156-clan)	`xpt` PP_5265

Accessory funnels that determine which branch carries flux: **adenine deaminase** (`PP_0591`, EC 3.5.4.2, adenine→hypoxanthine) and **guanine deaminase** (`guaD`/PP_4281, EC 3.5.4.3, guanine→xanthine). Their presence means salvage does not require a distinct enzyme per base — deaminated bases converge on the HGPRT/XPRT branches.

4. Candidate genes and evidence

Evidence tiers derived from UniProt UP000000556 (reviewed status, EC assignment, protein-existence).

Gene	Locus / Acc	Step	Evidence tier	Verdict	Curation notes
`apt`	PP_4266 / Q88F33	2 (Ade→AMP)	**Reviewed, EC 2.4.2.7, PF00156**	`covered`	Single-copy APRT; strong. Homology-inferred but unambiguous famil
`xpt`	PP_5265 / Q88CB6	4 (Xan→XMP)	**Reviewed, EC 2.4.2.22**	`covered`	XPRT. (No Pfam ma in UniProt — minor annotation quirk, n concern.)
`ppnP`	PP_4248 / Q88F51	1 (phosphorolysis)	**Reviewed, EC 2.4.2.1/2.4.2.2, PF06865 (Cupin)**	`covered` (accessory)	Genuine PpnP class broad pyrimidine+purine specificity (P 35094440 In *E. coli* PpnP is accessory to DeoD, s may be low-capacit here.
`hpt`	PP_0747 / Q88PV1	3 (Hx/Gua→IMP/GMP)	**Unreviewed, "Predicted", NO EC, PF00156**	`covered` / **`candidate_uncertain`**	Name-only HGPRT; broad PRT clan. Promote to full revi
`PP_3254`	Q88HU9	1 (phosphorolysis)	Unreviewed, "Predicted", **PF01048** / **IPR000845**	**new candidate** for step 1	Not in candidate lis Nucleoside-phosphorylase-dom (DeoD/MTAP/UP superfamily); best h for a classical high-f PNP. **Promote.**
`yfiH`	PP_0624 / Q88Q72	(claimed 1)	Unreviewed, **no EC**, PF02578 (DUF152/YfiH)	**likely over-annotation**	"Purine nucleoside phosphorylase" nar unsupported; do no score.

Gene	Locus / Acc	Step	Evidence tier	Verdict	Curation notes
`PP_3230`	Q88HX3	(possible 3/4)	Unreviewed, PF00156 "PRT-domain protein"	`candidate_uncertain`	Possible additional/ second PRT (e.g. a `gpt` -type). Check substrate.
`PP_0591` (Ade deaminase)	Q88QA3	funnel	Reviewed, EC 3.5.4.2	context	Routes adenine→hypoxant species-relevant flu nuance.
`guaD`	PP_4281 / Q88F18	funnel	Unreviewed, EC 3.5.4.3	context	Guanine→xanthine.
`amn`, `PP_3662`	PP_4779, PP_3662	(not salvage)	EC 3.2.2.4	out-of-scope	AMP nucleosidase = AMP→adenine+R5P (degradation, not salvage). `amn` shar PF01048 but is functionally distinc

Paralog ambiguity: The type-I PRT clan **PF00156** in KT2440 contains `apt`, `hpt`/PP_0747, `purF`, `pyrE` (OPRT), `upp`/`pyrR`, `comF`, plus uncharacterised **PP_3230** and PP_0361. Pfam membership alone cannot assign a substrate; substrate specificity for `hpt`/PP_0747 and PP_3230 rests on best-hit orthology, not experiment.

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5. Gaps, ambiguities, and likely over-annotations

1. **Step-1 phosphorylase identity — the principal gap.** No DeoD/XapA ortholog exists in the proteome (only `ppnP` carries EC 2.4.2.1). Either (a) PpnP is the physiological purine-nucleoside phosphorylase in KT2440, or (b) **PP_3254** (uncharacterised PF01048/IPR000845) is the real workhorse. This is the single most important item to resolve. Verdict: **`candidate_uncertain`** for step 1.
2. **`yfiH`/PP_0624 over-propagation.** YfiH/DUF152 is a polyphenol-oxidase/laccase-like family; the "purine nucleoside phosphorylase" label is EC-less and unreviewed. Recommend **down-weighting / re-annotation review**; do not treat as salvage evidence.

3. `hpt`/**PP_0747 under-characterised.** Unreviewed + "Predicted" + no EC. Functionally likely correct (conserved HGPRT synteny across *Pseudomonas*), but evidence is homology-only.
4. `ppnP` **broad EC.** EC 2.4.2.1 **and** 2.4.2.2 with 8 substrate synonyms (adenosine…xanthosine, plus pyrimidines). This breadth is a family-level over-listing; in vivo purine contribution is uncertain.
5. **Catabolism mislabelled as core purine map.** `paoABC` (aldehyde oxidoreductase) and the ureide genes (`pucL/pucM/puuE/allA/allE`, `ureABC`, `xdhAB`) are catabolic/N-assimilatory, not salvage. Their inclusion inflates apparent module coverage — a `module_needs_revision` boundary issue.
6. `arcC`, `cysD/cysNC`, `pgm/cpsG/algC`, `cyaA`, `pde`, `relA/spoT`, `ppx`, **Nudix set** — clearly out of scope; over-inclusive projection from the overview map.

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6. Module and GO-curation recommendations

Per-step module verdicts: - Step 1 (nucleoside phosphorolysis): **`candidate_uncertain`** — PpnP present (EC-supported) but likely accessory; classical PNP unrepresented; investigate PP_3254. - Step 2 (adenine→AMP, `apt`): **`covered`**. - Step 3 (Hx/Gua→IMP/GMP, `hpt`): **`covered`** with a `candidate_uncertain` evidence flag (predicted, no EC). - Step 4 (xanthine→XMP, `xpt`): **`covered`**.

Boundary / module-document recommendations: - **`module_needs_revision` at the source:** the KEGG-`ppu00230`→module projection is far too broad. The salvage module document should enumerate **only** the 4 steps and explicitly exclude de novo synthesis, interconversion, catabolism, signalling and housekeeping hydrolases. - **Create a separate "purine catabolism / ureide N-assimilation" module** for `xdhAB`, `pucL/pucM/puuE/allA/allE`, `PP_4310`, `ureABC` (and possibly `paoABC` after re-annotation). This is where the bulk of the mis-attributed genes belong and is biologically important in *P. putida* (purines/allantoin as N sources). - **GO annotation targets:** `apt` → GO:0006168 (adenine salvage); `hpt` → GO:0032264 (IMP salvage) + GO:0032263 (GMP salvage); `xpt` → GO:0032265 (XMP salvage, if present) under GO:0043101 (purine-containing compound salvage); `ppnP`/PP_3254 → GO:0006152/GO:0006157-type purine ribonucleoside catabolic/phosphorylase activity (GO:0004731 purine-nucleoside phosphorylase activity). No new GO **term** requests appear necessary; existing salvage terms suffice.

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7. Genes to promote to full `fetch-gene` review

1. **PP_3254 (Q88HU9)** — *highest priority.* Test whether this IPR000845 nucleoside-phosphorylase-domain protein is the classical (DeoD/MTAP-type) purine nucleoside phosphorylase; would close the step-1 gap.
2. **`hpt` / PP_0747 (Q88PV1)** — confirm HGPRT substrate range and assign EC 2.4.2.8; currently predicted/no-EC.
3. **`yfiH` / PP_0624 (Q88Q72)** — adjudicate the "purine nucleoside phosphorylase" name vs. the YfiH/DUF152 polyphenol-oxidase assignment; likely re-annotate.
4. **PP_3230 (Q88HX3)** — resolve whether this extra PF00156 PRT is a second hypoxanthine/xanthine-guanine PRT (`gpt`) or unrelated.
5. **`ppnP` / PP_4248 (Q88F51)** — confirm physiological purine (vs pyrimidine) role and whether it is the sole phosphorylase.

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8. Evidence status and open questions

- **Direct target-organism experimental evidence:** none located for the salvage enzymes themselves (KT2440 salvage biochemistry is largely unstudied). Available KT2440 literature retrieved concerns unrelated topics (biofilm/(p)ppGpp, malonyl-CoA engineering).
- **Homology / database-inferred (this review):** all four PRT/phosphorylase assignments are homology-based; the two SwissProt-reviewed entries (`apt`, `xpt`) carry the strongest inference, `ppnP` is reviewed, `hpt` is predicted-only. Proteome composition (presence of PpnP, absence of DeoD/XapA, existence of PP_3254) is **directly verified** against UniProt UP000000556 (this work).
- **Cross-organism transfer:** DeoD-as-workhorse and PpnP-as-accessory are *E. coli* facts (P 17639373; P 35094440); transfer to KT2440 is **weak-to-uncertain** precisely because KT2440 lacks DeoD — so the *E. coli* paradigm cannot be assumed.
- **Resolving experiments:** (i) enzymatic assay of purified PP_3254 and PpnP on inosine/guanosine/adenosine; (ii) growth of Δ*ppnP*, ΔPP_3254 and Δ*hpt*/Δ*apt*/Δ*xpt* mutants on the

corresponding nucleosides/bases as C/N source; (iii) substrate profiling of PP_0747 and PP_3230 to settle the HGPRT/XPRT/GPT split.

Key references

- Wen et al. 2022, *Int J Biol Macromol* — Crystal structures of PpnP (pyrimidine/purine nucleoside phosphorylase), a distinct Cupin-fold NP class that phosphorolyses diverse nucleosides to ribose-1-phosphate + free base.

  P 35094440

- Modrak-Wójcik et al. 2008, *Eur Biophys J* — *E. coli* purine nucleoside phosphorylase PNP-I is the product of the *deoD* gene (reference for the canonical workhorse absent in KT2440).

  P 17639373

- UniProt Knowledgebase, proteome **UP000000556** (*P. putida* KT2440) — entries Q88F51 (`ppnP`), Q88F33 (`apt`), Q88CB6 (`xpt`), Q88PV1 (`hpt`/PP_0747), Q88Q72 (`yfiH`), Q88HU9 (PP_3254), Q88HX3 (PP_3230); queried this work.
- KEGG map **ppu00230** "Purine metabolism" (overview-map scope caveat).
